

	Hence unit velocity is halved.
6(a)(i)	$V_{out} = IR = (2.7 \times 10^{-3})(90) = 2.43 \text{ V}$
6(a)(ii)	Effective resistance of thermistor and resistor Q $= \left(\frac{1}{120} + \frac{1}{90} \right)^{-1}$ $= 51.429 \text{ } \Omega$ $\frac{90}{90 + R_p} \times 9.0 = 2.43$ $R_p = 243.3 \text{ } \Omega$ $V_{out} = \frac{51.429}{51.429 + 243.3} \times 9.0$ $= 1.57 \text{ V}$
6(a)(iii)	When temperature of thermistor increases, its resistance drops. Since thermistor and resistor Q are connected in parallel, this will cause the effective resistance across them to decrease. By potential divider principle, V_{out} decreases as resistance of P remains unchanged.
6(b)	Length AB $= r\theta$ $= (1.2) \left(\frac{92}{180} \times \pi \right)$ $= 1.927 \text{ cm}$ Since resistance of uniform wire is directly proportional to its length, By potential divider principle, $V_{out} = \frac{1.927}{6.5} \times 9.0$ $= 2.7 \text{ V}$

7(a)	$V = kf^n$, where k and n are constants
7(b)	